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A class of sp^3 boron-based single-ion polymeric electrolytes for lithium ion batteries†

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A class of sp^3 boron-based polymeric compounds with highly exposed lithium cations is both thermally and electrochemically stable for use in Li-ion batteries. Syntheses of a variety of the sp^3 boron-based polymeric materials can be realized by taking advantage of the well-developed methods used for preparation of metal organic framework compounds.

It has become increasingly evident that current technologies of lithium ion rechargeable batteries would not be able to provide sufficient electricity required to power electric vehicles with a driving range of 300 miles.[1,2] While significant development has been made to enhance energy capacity of battery devices by utilizing new or performance-enhanced electrode materials,[3–5] much less effort has been devoted to boosting battery performance by employing new electrolytes. It is true that the energy capacity of batteries is largely determined by the electrode materials; however, an efficient electrolyte can play a critical role in enabling an optimal selection of pair of cathode and anode and a better device design to maximize the performance of batteries.[6–8] Single ion conducting polymer electrolytes offer several unique advantages over the widely used electrolytes made of small molecular lithium salts dissolved in organic solvents as well as the conventional solid polymeric electrolytes. Lithium ion transport can be substantially improved in single ion polymer electrolytes than in dual-ion electrolytes because the anions are covalently bonded to the polymers and thus immobilized in the charge-discharge processes. Consequently, the polarization effects on the electrodes are minimized,[9] facilitating more complete redox reaction in the battery cells. Furthermore, single ion polymer electrolytes made in membrane forms with appropriate film uniformity, density, thickness and mechanical strength can be directly utilized as solid battery separators. These membrane-based separators may potentially enable a lithium foil to be used as an anode directly to maximize the anode capacity and cell voltage by blocking formation of Li dendrites.[10–12] Finally, the use of single-ion polymer electrolytes is also expected to help improve battery safety compared to the liquid electrolytes of small lithium salts in organic solvents.[12,13] Unfortunately, to date, reports on single ion polymer electrolytes with ionic conductivity in Li-ion batteries higher than 10^{-4} S cm^{-1}, a wide electrochemical window and good thermal stability required for high power applications have been rare. In this Communication, we report synthesis and membrane fabrication of a class of sp^3 boron-based polymeric materials with tunable one dimensional and three dimensional networks that exhibit excellent ionic conductivity and high thermal and electrochemical stability. Each sp^3 boron atom is covalently bonded to four oxygen atoms that are associated with electron withdrawing groups *via* aromatic π-conjugation in the polymer backbones. This structural arrangement allows the ionized charges from lithium atoms to be redistributed to the aromatic rings, leading to the enhanced mobility of lithium ions in the extra-framework. Therefore, the mechanism of Li-ion transport is fundamentally different from the conventional solid electrolytes, which rely on liquid electrolytes of small lithium salts to transport ions in the polymer matrix and thus still remain to be dual-ion electrolytes in nature.

The idea of making use of sp^3 boron for the synthesis of lithium salts as electrolytes upon dissolution in organic solvents for Li-ion batteries is not new. In fact, several sp^3 boron-based liquid electrolytes have already been reported, some of which display remarkable thermal and electrochemical stability.[14,15] The charge delocalization in the large anions of the salts gives rise to high ionic conductivity in the selected organic solvents. Nevertheless, the liquid electrolytes may suffer from potential safety risks caused by leakage or exposure to air or by combustion of organic solvents upon overheating resulting from short circuiting between cathode and anode.[13] In addition, liquid electrolytes are generally not suited for Li–S and Na–S batteries.[16,17] The problems associated with liquid electrolytes can be largely overcome by constructing an extended polymeric structure that contains locally well-distributed anionic species either in the

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backbones or in the side chains. Furthermore, the electrolytes can be made in the form of sol–gel to enhance the ionic conductivity, which requires interconnected cavities in the polymer matrix so that organic solvents can be accommodated. Therefore, a rational approach is to construct a polymeric network with micropores, ideally, similar to those found in metal organic framework (MOF) compounds. However, unlike most of the MOF materials, the lithium ions in the sp^3 boron polymeric compounds are located in the extra-framework and loosely bound to the anionic sites *via* electrostatic interactions, which would thus give rise to high ion mobility in organic solvents. Therefore, an ideal single ion polymer electrolyte should have the two important attributes: (1) highly exposed cations with charge-delocalized anions and (2) adequate pores to allow infusion of organic solvents to make polymer gel for conductivity enhancement. A handful of single ion electrolytes have recently been reported.[10,11,18–20] Unfortunately, the reported ion conductivity has remained to be relatively low for high power applications.

Here, we present three prototypical sp^3 boron-based polymeric materials with the structures shown in Fig. 1, representing 1-dimensional aromatic, 3-dimensional aromatic and 1-dimensional aliphatic networks, respectively. The propagation of polymerization was designed based on the target structures using the precursors typical for the synthesis of MOF compounds. These polymeric materials represent different dimensionality and rigidity. The synthesis of the target materials includes two steps (Scheme 1): (1) silylation of pyromellitic acid, terephthalic acid and 1, 2, 3, 4-butanetetracarboxylic acid and (2) reaction between the silylation derivatives and lithium tetramethanolatoborate ($LiB(OCH_3)_4$) to produce the final products.

The FE-SEM images and the X-ray diffraction patterns of PPAB, PTAB and PBAB are shown in Fig. 2. Here, PPAB displays a cube-like morphology with a length approximately 1 μm; the PTAB particles exhibit a cube-like shape with a dimension of 1 μm and PBAB particles are with a flake shape of a length about 1 μm × 0.5 μm. The morphology of both PPAB and PTAB appears to be quite sharp due to the high rigidity of the aromatic precursors, while the morphology of PBAB is "softer", reflecting the flexibility of the aliphatic precursor. The XRD patterns of the three compounds

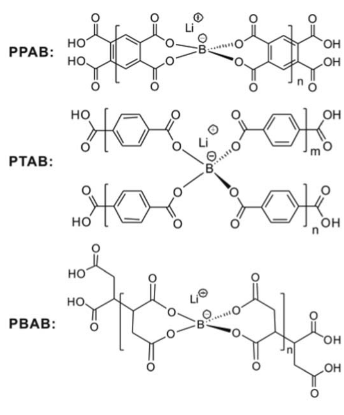
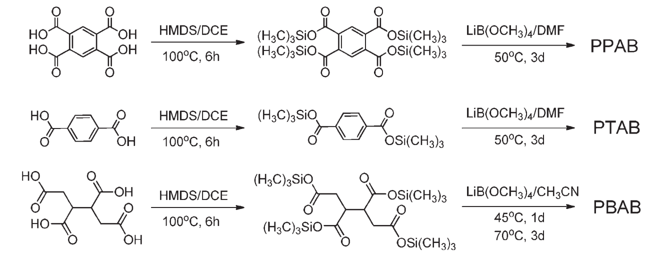


Fig. 1 The structures of the three sp^3 boron-based polymeric compounds.



Scheme 1 Procedure for the synthesis of the target compounds.

display sharp diffraction peaks, suggesting that these materials have a long range order.

Fig. 3 depicts the thermal behavior of PPAB, PTAB and PBAB using thermal gravimetric analysis (TGA) under nitrogen atmosphere. The results indicate that these compounds start to decompose at 210 °C, 570 °C and 250 °C, respectively. The small amount of weight loss of PPAB before 100 °C is attributed to the desorption of water molecules adsorbed upon brief exposure to air. The TGA spectra suggest that these materials surpass the thermal stability requirement for electrolytes in Li-ion batteries and are thus well suited for applications in a high temperature range.

The electrochemical stability of PPAB, PTAB and PBAB was tested using linear sweep voltammetry (LSV). Prior to the test, three gel polymer composite solutions were prepared separately by dissolving the mixtures of PVDF and the polymer composites with the weight ratio of 3 : 1 in an EC/PC solution with the volume ratio of 1 : 1 at high temperature in the glove box. The gel polymer membranes were fabricated *via* cooling. Subsequently, the gel polymer electrolyte membranes (GPEMs) were used to construct Li|GPEMs|stainless steels (SS) cells. The current–voltage response of these membranes tested with the Li|GPEMs|SS cells at room temperature is shown in Fig. 4. Clearly, the membranes of PPAB, PTAB and PBAB are stable electrochemically up to 5.2 V, 4.2 V and 4.3 V, respectively, providing a nice electrochemical window for

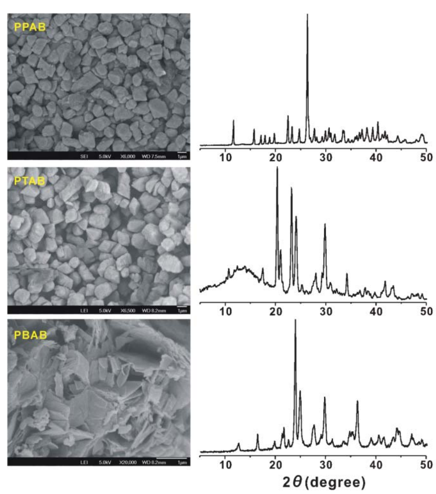


Fig. 2 The SEM images and the XRD spectra of PPAB, PTAB and PBAB.

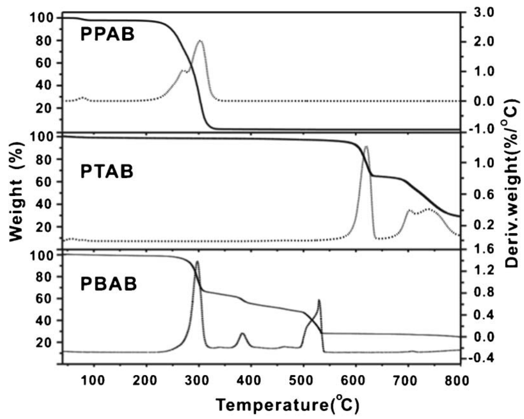


Fig. 3 TGA graphs and derivative curves of PPAB, PTAB and PBAB.

high-voltage Li-ion battery applications. The ability of an electrolyte to sustain a high voltage is of particular importance for a cathode made of multi-component electrode materials, such as $LiCu_xMn_yO_4$[21] and $LiCo_xMn_yO_4$,[22] which have been shown to have significantly higher energy capacity than the conventional cathode materials such as $LiFePO_4$ and $LiMn_2O_4$. In particular, the high electrochemical stability of PPAB up to 5.2 V is well-suited for use of multi-component oxides as cathode for which most commonly used polymer electrolytes are unable to meet the high voltage requirement ($\sim$4.9 V).

Finally, we utilized electrochemical impedance spectroscopy (EIS) to determine the conductivity of the above GPEMs in the SS|GPEM|SS cells and the Nyquist plots are shown in Fig. S3, ESI.† The conductivity of the GPEMs was derived to be 1.26, 1.78 and 5.81 mS cm^{-1} at room temperature, respectively. These values are significantly higher than the reported values of gel polymer electrolytes[19,23] and close to the value of the commercial liquid electrolyte ($LiPF_6$, close to 10 mS cm^{-1}).[24,25] With a thickness of roughly 100 μm and good mechanical strength, the membranes can serve both as electrolytes and as separators for Li-ion batteries. The high ionic conductivity would enable the batteries to use polymer electrolytes with performance comparable to the conventional liquid electrolyte but with better safety. At the same time, the mechanically strong membranes may prevent lithium dendrite formation when a lithium foil is used as an anode.

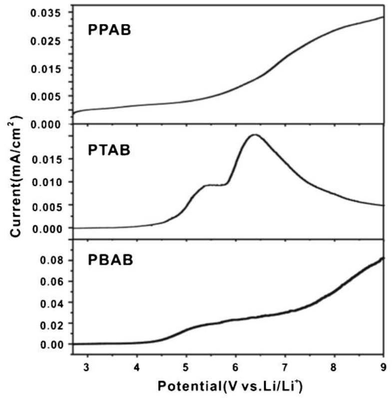


Fig. 4 Electrochemical stability of PPAB, PTAB and PBAB.

In summary, a new class of sp^3 boron-based framework compounds has been successfully developed. These materials exhibit good thermal and electrochemical stability and are well suited for applications in Li-ion batteries as electrolytes as well as separators. Membranes fabricated with these compounds exhibit wide electrochemical windows; the ionic conductivity of the membranes is in the same order of magnitude as the conventional liquid electrolytes. The sp^3 boron-based compounds can be made using the strategy widely utilized for MOF syntheses with not only nano or micro size pores but, more importantly, a strong functionality arising from the highly exposed lithium cations weakly bonded to the charge-delocalized anions in polymer networks. The large pores enable organic solvents to readily infuse into the polymer matrix to make gel electrolytes without solvent leakage upon charging and discharging, which thus provide an easy passage for the highly exposed lithium ions to be transported rapidly between electrodes with conductivity comparable to the conventional liquid electrolytes. Therefore, one can take advantage of the widely available MOF precursors to make a variety of new sp^3 boron-based framework compounds.

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